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Comparison and analysis of deep learning for Dehazing

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***Abstract*:** Hazy image is caused by environmental factors such as uneven air light, darkness, contrast, saturation, and attenuation. Due to limited visibility, various applications such as ITS, tracking systems, object detection devices, and others may fail. Image dehazing is used to minimise the effects of weather conditions and improve the pictorial impacts of the image which make post-processing easier. Various approaches are utilized to create high-quality images with improved colour, lighting, and boundaries which are free of halo artifacts. To eliminate the haze, most known de-hazing algorithms employ a conventional scattering model, prior's methods. This research compares a current de-hazing method based on deep learning to existing state-of-the-art technologies.

Keywords— Deep learning, Haze, CNN, Dark Channel Prior, Dehazed

I. Introduction

Absorption or scattering of light travelling through water droplets floating in the air produces haze. The aim of de-hazing methods is to repair hazed image into a clear haze free image, because various progressive image processing need a haze free image [1]– [6]. Methods for hazy image processing can be categorized as: One is enhancement method and second is based on physical model. Enhancement based method solve the purpose of clarity. When there are more differences in scene depth and the image scene needs to be de-hazed, this method does not perform well. Enhancement techniques are Bi-histogram modification [7], Histogram modification [8] Histogram Equalization [9], Weighted Histograms [10], etc. These techniques suffer from saturation of pixels like over and under enhancement. Physical model based algorithms [10], [11] are used for precisely approximation of global atmospheric light α and transmission map t(x). And by using model of atmospheric scattering,clear images can be recovered. On complicated scenes, this model has a better effect, and image information is saved more thoroughly. The atmospheric scattering model defined as [12]- [13]:

$$f(x) = I(x)\, t(x) + \alpha\,(1 - t(x)) \qquad (1)$$

In Equation. (1) $f(x)$ called as hazy image, $I(x)$ is original haze-freescene, $t(x)$ is termed as scene transmission and α is the global atmospheric light. The terms $t(x)$ andα terms are not given. This is the reason that prior method for image dehazing algorithms need to calculate transmission map andα and after this recover the de hazed image using Equation 1. Even though numerous methods have produced acceptable results, they are supported by strong assumption and need a number of image generation parameters. The estimation of t(x) is generally inaccurate when color of scene is similar with the haze which affects the performance of de-hazing algorithms [14], [15]. So, the traditional haze removal algorithms are not hold good in some conditions. Therefore, development in deep learning, convolution neural network is used for de-hazing [16]– [17]. Due to good performance of CNN in computer vision application it has been used for image de-hazing. Such techniques can learn not just too difficult input-output interactions autonomously from data annotations, and also more complex heuristics learning that humans are unable to identify.

Convolution neural network based approaches are more robust than classic non-learning based methods. Convolution neural network based methods used atmospheric scattering model defined by Equation (1) use nonlinear mapping input image and transmission map. Ren et al. [18] proposed coarse network to build coarse-grained transmission maps and used refine image with fine networks. But, this method need manual parameter adjustments based on the density of haze in image. Cai et al. [19] also used CNN to learn the features of hazy image and to estimate transmission map number of molecular clue have been used. But transmission map under some specific circumstances are in accurate and de-hazing impact is inadequate. Li et al. [20], presented atmospheric scattering model into the network by incorporating a light-weight CNN. This method helps to minimize the recovery errors .and improve the image quality.

This paper is classified into sections. Section II defines the review linked to the research in de-hazing using enhancement based methods, physical restoration algorithms and deep learning algorithms. Section III defines the comprehensive experiments results. Section IV is conclusion.

II. Related Work

Numbers of algorithms have been suggested to solve image de-hazing problem in computer vision applications. Methods for hazy image processing can be classified into following types. One is enhancement based and second is based on physical model and third is learning-based algorithms. Main transformation between learning methods and prior-based methods is that the prior methods extract the features by handcrafted methods, whereas learning method extracts the features inevitably by Convolution neural network. Above

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two methods have their own benefits and drawbacks. Section II, focus on image de-hazing methods learning-methods and prior- methods and discuss their benefits and drawbacks.

Model of Atmospheric Scattering

Equation. (1), can be represented as:

$$I(x) = \frac{f(x) - \alpha}{t(x)} + \alpha \qquad (2)$$

The terms α, I (x) and t (x) are not known. As shown in Equation. (2), the actual scene I (x) get recovered effortlessly after the calculation of t(x) and α. Equation. (2) indicates that t(x) and α are the terms required for atmospheric scattering model based image de-hazing methods. As a result, the majority of classic atmospheric scattering model-based methods are focused on precisely computing the global atmospheric light α and transmission t(x). Transmission scene t(x) can be defined as:

$$t(x) = e^{-\beta d(x)} \qquad (3)$$

β is known as coefficient of scattering.

Prior-Based De-Hazing Algorithms

Methods for eliminating haze using single image must be based on set of assumptions. Whereas assumptions can be application-based, heuristic based hypotheses, Statistical characteristics based. But the problem in model-based de-hazing strategies is to accurately estimate the model parameters, such as the transmission map and global ambient light. In. [21], author presented the work based on dark channel that was used in many prominent model-based dehazing techniques.

Dark Channel Prior

For outdoor clear images, few pixels are observed as low intensity pixels or near to zero in one-color channel exist in most patches that are not covered by the sky. The channel that has very low intensity pixels or close to zero are called as dark pixels. There are a number of reasons that contribute to dark pixels [9]:

- The dark pixel is caused by the image's shadow.
- Colorful things are responsible for the dark pixels. Flowers, automobiles, buildings, road signs, and leaving are just a few examples
- Black things such as road signs, tree trunks, and vehicle tyres are responsible for the dark pixels. [21].

If an image patch contains at least one of these elements, the patch must include dark pixels. The pixel with low intensity in any of the channels (RGB) and below the threshold value is called as a dark pixel.

$$I(x) \text{ is value of dark pixel} = min_{c\in\{r,g,b\}} Ic(x) <= \delta \quad \textbf{(4)}$$

Every pixel in I^{dark} is a scalar term. DC is the result of two minimum operators as shown in equation (7). One operator $min_{ce\{r, g, b\}}$ is performed on each pixel. Operator $min_{x' \varepsilon\Omega(x')}$ is a minimum filter. The I^{dark} of a hazy image I can be computed by

$$I^{dark} = min_{m' n' \varepsilon\Omega(x')} min_{ce\{r, g, b\}} (I_c(x') \qquad (5)$$

The pixel value of dark channel can be represented as: $I^{dark} =\sim 0$ this observation is called as Dark Channel Prior.

Enhancement Based Dehazing

Image enhancement process aims to remove the existence of noise and enhance on the information gathered continuously to produce high quality image, using this sort of method. On hazy images, histogram equalization is a common process. Instead of compensating for the physical model of image degradation, this approach boosts the contrast of an image by adjusting grey scale values throughout a predefined spectrum that is equally distributed. Local Histogram Equalization and Global Histogram Equalization are the two types of histogram equalization.

CAP (COLOR ATTENUATION PRIOR)

In the work given in [22], author presented the CAP to find various features of image in different hazy conditions. Author found that haze concentration affects the saturation and intensity of hazy images. To be accurate, the saturation is fairly high in a less-affected zone with a hazy image, and the brightness is within the typical range. But, for certain area where image is affected by heavy haze, its intensity will increase abruptly, and saturation will decay. Hence, the difference between saturation and the brightness becomes greater. So, Zhu et al. used following equation:

$$Z(x) \propto h(x) \propto m(x) - c(x) \qquad (6)$$

Here, h(x) is the haze concentration, z(x) is the scene depth, the m(x) and c(x) are the brightness and saturation respectively. Hence, the depth of the hazed image can be assessed by the CAP.

Learning-Based Algorithms

Researchers have focused on deep learning to examine how good it performs on image haze removal, as the learning

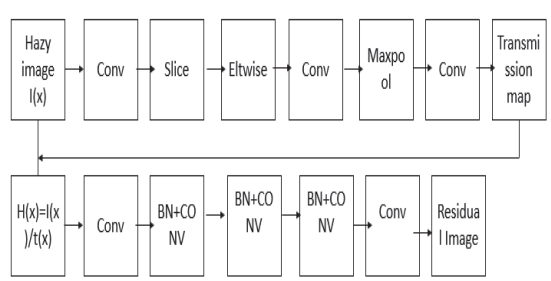

Figure1. Network structure for residual-based deep CNN

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based algorithms gives extraordinary outcomes of CNN in high-level vision tasks such as image classification. In order to solve image inverse difficulties, DNN have been introduced. Using the nonlinearity and enormous capacity of

DNN, deep learning algorithms straightly learn a mapping from the corrupted images to the high quality image. Different organizations have been contrived for image super-goal, medical image recovery, image denoising, in painting. For mapping to the target high-quality image, all of these deep learning-based approaches rely on a black-box deep neural network. These networks designs comprise of regularly used organization layers, like convolution, ReLU, cluster standardization. These network layers are generic layers that aren't developed with domain knowledge or envisioning models in mind for solving image inverse challenges. Due to this constraint, learning the network to effectively map input degraded picture / samples to the desired high quality images requires a big training dataset and appropriate training methods.

Residual-Based Deep CNN

In the work given [23], author presented an algorithm where only transmission map is required to estimate by a network whereas residual network is required to achieve the haze free image. In this algorithm estimation of transmission map is done by a network which is consisting of six layers as shown in the Figure 1. The first layer which is convolution layer extracts the feature from input image by using convolution kernel of size 3 and obtains the feature map. Following Equation can be used to determine the features of the convolutional output.

$$Conv=f\,(y \times M + n).$$

Where y represents the image matrix, M denotes the convolution kernel, × represents operator, and n denotes the offset value. f denotes the ReLU function.

The 2nd layer is slice layer and function of this layer is to slice the previous obtained feature map and covert the size which is used for next layer for processing. The 3rd is Eltwise layer which performs three actions that provides the max, sum, product values. The 4th layer is uses multi-level mapping to enhance the accuracy of Feature extraction. Next 5th layer performs extraction on the input image using pixel-level neighbor maximum criteria using convolution kernel, that helps to maintain texture features, overcome feature map local sensitivity, and reduce propagation map error. The 6th layer is a conv layer that translates the input image data into a feature map and uses the ReLu activation layer to modify the feature map nonlinearly. And finally to get the dehazed image output of last layer is combined with residual network. In proposed method DnCNNs network and ResNet network as a residual network. This model effectively performs well on de-hazing for different scenes, and has no color alteration.

Deep Residual Haze Network

C.Wang et.al [24], proposed residual, deep learning method for single image de-hazing. This proposed method does not require α and transmission estimation t(x). DRHNet works on the principle to perform subtraction of the input hazed image from the learned residual map. DRHNet method has three components as shown in Fig 2. The first component is used to find the cumulative contextual information and encodes into feature maps, DRHNet method a context-aware feature extraction module. In this method α and t(x) are not required to obtain the de-hazed image.

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Perceptual Pyramid Deep Network

He Zhang et.al [25], presents CNN based multi-scale image de hazing algorithms. The proposed method used the encoder and decoder architecture to get the dehazed image. The encoder is required to take input image and maps its intermediate features and decoder is used to map latent space of clear image. Encoder is consisting of DnCNN and residual blocks and each block is combination of layers where every layer get the feature map as input quantity from the previous one and this process ensures the maximum information in both forward and backward and makes the training process easy. The decoder is consisting of dense block, up sampling block & residual blocks. This combination recovers the high quality details that lost during the encoding process and provides high quality reconstruction of de-hazed image. The proposed method directly maps the nonlinear mapping between hazy and clear image and used perceptual LF to train the network and reduced the blur in the output image.

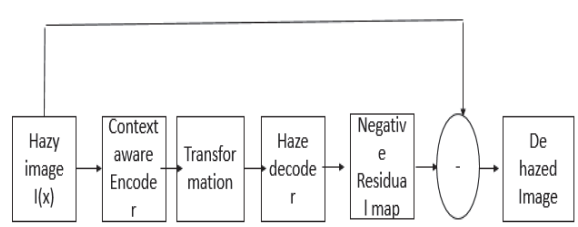

Figure 2. Network structure for Deep Residual Haze Network

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DE-HAZE NET

Cai et al. [26] proposed single image de-hazing using De-Haze Net that is the inversion of Koschmieder model. It presents a innovative method for calculating the t(x) transmission map. De-haze Net calculates transmission map

t(x) of hazed image and output its consistent transmission map, which is required to recuperate a de-hazed image using inversion of the model. Thousands of hazy patches, generated from haze-free images, have been used to train DehazeNet. A ground truth for the estimation of transmission map can be providing for training because the mechanism for creating these hazy patches is known. In this proposed method atmospheric light is calculated separately.

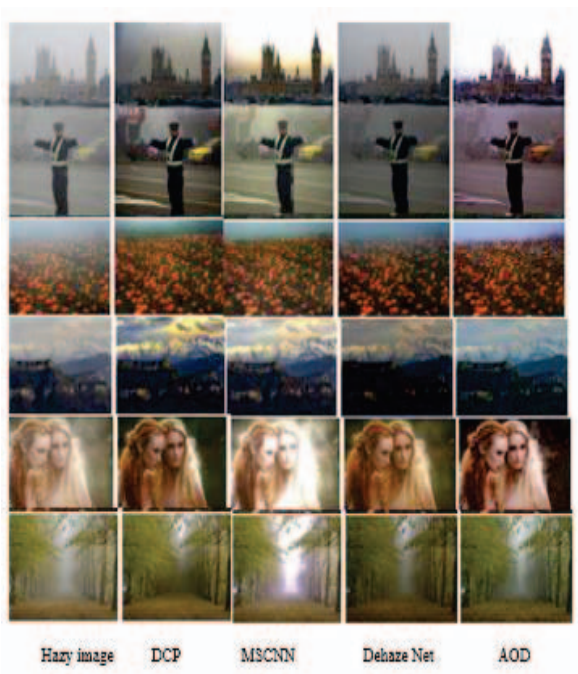

Figure 3. Qualitative comparison of results

III. Experiments

The goal of our research is to compare deep learning-based haze removal algorithms like Residual-Based Deep CNN, Deep Residual Haze Network, DehazeNet and classic haze removal methods. To conduct the comparison, we worked on dataset of NYU-Depth V2 for synthetic training images. The New York University Depth V2 dataset includes 1,449 heavily annotated indoor color photos with ground truth depth information. To fill in lacking depth labels, the raw depth map was projected and colored. In this comparison, we employed quality metrics like PSNR and SSIM. Learning-based approaches like, AOD, and DehazeNet generate significantly improved PSNR and SSIM outcomes than model-based methods. Deep residue learning and fixed-point iterations, both contribute to considerable performance gain.

IV. Evaluation Standard

A decent de-hazing method does not only improve texture information visibility and edge, but also keep the image's structure and color in good condition. Most commonly used evaluating objective and assessing quality of image is PSNR. For two images Iin and Iout, the PSNR between these two is defined as [26]:

$$PSNR = 10\mathrm{Log}_{10}\frac{255}{\sqrt{Iin^2+Iout^2}} \qquad (9)$$

Where Iin is the image free from haze and Iout is resultant de-hazed image. If the Peak signal to noise ratio value is higher, distortion is less and image quality is good. SSIM is found to be associated with the HVS's perceived quality. It calculates image similarity based on brightness, contrast, and structure. SSIM can be defined as:

$$\mathrm{SSIM(xin,xout)}=\left(\frac{2\mu_{Xin}\mu_{X}out+\theta 1)(2\sigma_{Xinxout}+\theta 2)}{(\mu 2_{Xin}+\mu 2_{Xout}+\theta 1)(\sigma^2_{Xin}+\sigma^2_{Xout}+\theta 2)}\right. \qquad (10)$$

SSIM value must be lie in in between 0 to 1. Larger the value of SSIM, distortion is less.

TABLE 1. Average PSNR and SSIM result

Metrics	DCP	MSCNN	Dehaze Net	AOD
PSNR(db)	16.64	18.78	15.20	14.02
SSIM	0.82	0.83	0.74	0.81

V. Conclusion

Image de-hazing strategies are rapidly increasing research subject in the field of machine vision, with an incredible number of new strategies appearing all the time. The current de-hazing approaches are observed in this work from a different angle, including mathematical, theoretical models and performance metrics. Deep CNN has been demonstrated to be an effective solution for image de-hazing in recent studies. Nonetheless, we may forecast that deep learning will eventually provide the best possible recovery of model parameters. AOD has indeed made significant progress in the estimation of air light, and if fine-tuned for practical systems, it may become efficient. Those approaches are still limited by the imaging model, and further research into the reformulation of the haze model is needed to achieve a revolution in performance

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REFERENCES

[1] C.Tian,Y.Xu,andW.Zuo,''ImagedenoisingusingdeepCNNwithbatch renormalization,'' Neural Netw., vol. 121, pp. 461–473, Jan. 2020.

[2] J.Ker,L.Wang,J.Rao,andT.Lim,''Deeplearningapplicationsinmedical image analysis,'' IEEE Access, vol. 6, pp. 9375–9389, 2018.

[3] J. Zhang, B. Chen, M. Zhou, H. Lan, and F. Gao, ''Photoacoustic image classificationandsegmentationofbreastcancer:Afeasibilitystudy,''IEEE Access, vol. 7, pp. 5457–5466, 2019.

[4] C. Zhou, H. Fan, and Z. Li, ''Tonguenet: Accurate localization and segmentation for tongue images using deep neural networks,'' IEEE Access, vol. 7, pp. 148779–148789, 2019.

[5] Z. Li, G. Liu, D. Zhang, and Y. Xu, ''Robust single-object image segmentation based on salient transition region,'' Pattern Recognit., vol. 52, pp. 317–331, Apr. 2016.

[6] H. Fan, F. Zhang, L. Xi, Z. Li, G. Liu, and Y. Xu, ''LeukocyteMask: Anautomatedlocalizationandsegmentationmethodforleukocyteinblood smearimagesusingdeepneuralnetworks,''J.Biophotonics,vol.12,no.7, 2019, Art. no. e201800488.

[7] Chen BH, Huang SC, Ye JH (2015) Hazy image restoration by bi-histogram modification. ACM Tran Intell Syst Technol TIST 6(4):50 K. Elissa,

[8] Ghani ASA, Isa NAM (2017) Automatic system for improving underwater image contrast and color through recursive adaptive histogram modification. Comput Electron Agric 141:181–195

[9] Fu X, Wang J, Zeng D, Huang Y, Ding X (2015a) Remote sensing image enhancement using regularized-histogram equalization and dct. IEEE Geosci Remote Sens Lett 12(11):2301–2305 M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.

[10] He S, Yang Q, Lau RW, Yang MH (2016) Fast weighted histograms for bilateral filtering and nearest neighbor searching. IEEE Trans Circ Syst Video Technol 26(5):891–902

[11] K. He, J. Sun, and X. Tang, ''Single image haze removal using dark channel prior,'' IEEE Trans. Pattern Anal. Mach. Intell., vol. 33, no. 12, pp. 2341–2353, Dec. 2011

[12] Q. Zhu, J. Mai, and L. Shao, ''A fast single image haze removal algorithm singcolorattenuationprior,''IEEETrans.ImageProcess.,vol.24,no.11, pp. 3522–3533, Nov. 2015

[13] W. Ren, S. Liu, H. Zhang, J. Pan, X. Cao, and M.-H. Yang, ''Single image dehazing via multi-scale convolutional neural networks,'' in Proc. IEEE Conf. Comput. Vis., Sep. 2016, pp. 154–169.

[14] B. Cai, X. Xu, K. Jia, C. Qing, and D. Tao, ''DehazeNet: An end-toend system for single image haze removal,'' IEEE Trans. Image Process., vol. 25, no. 11, pp. 5187–5198, Nov. 2016.

[15] L. Liu, B. Liu, H. Huang, and A. C. Bovik, ''No-reference image quality assessmentbasedonspatialandspectralentropies,''SignalProcess.,Image Commun., vol. 29, no. 8, pp. 856–863, 2014.

[16] Q. Zhu, J. Mai, and L. Shao, ''A fast single image haze removal algorithm usingcolorattenuationprior,''IEEETrans.ImageProcess.,vol.24,no.11, pp. 3522–3533, Nov. 2015.

[17] B. Cai, X. Xu, K. Jia, C. Qing, and D. Tao. Dehazenet: An end-to-end system for single image haze removal. IEEE Transactions on Image Processing, 25(11):5187–5198, 2016

[18] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In Proceedings of the IEEE conference on CVPR, pages 770–778, 2016

[19] C.Wang et.al, Deep Residual Haze Network for Image Dehazing and Deraining IEEE Access, 2020.

[20] B. Li, X. Peng, and Z. Wang, J. Xu, and D. Feng, ''AOD-Net: All-inOne dehazing network,'' in Proc. IEEE Conf. Comput. Vis., Oct. 2017, pp. 4780–4788.

[21] S.IoffeandC.Szegedy,''Batchnormalization:Acceleratingdeepnetwork training by reducing internal covariate shift,'' in Proc. IEEE Conf. Mach. Learn., Mar. 2015, pp. 448–456.

[22] K. He, X. Zhang, and S. Ren, and J. Sun, ''Deep residual learning for image recognition,'' in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., Jun. 2016, pp. 770–778.

[23] Y. Song et al., ''CREST: Convolutional residual learning for visual tracking,'' in Proc. IEEE Conf. Comput. Vis., Oct. 2017, pp. 2574–2583.

[24] T. Pohlen, A. Hermans, and M. Mathias, and B. Leibe, ''Full-resolution residual networks for semantic segmentation in street scenes,'' in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., Jul. 2016, pp. 3309–3318.

[25] K. He, X. Zhang, S. Ren, and J. Sun, ''Identity mappings in deep residual networks,'' in Proc. Eur. Conf. Comput. Vis., 2016, pp. 630–645.

[26] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, ''Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising,'' IEEE Trans. Image Process., vol. 26, no. 7, pp. 3142–3155, Jul. 2016.

[27] C. Chen, M. N. Do, and J. Wang, ''Robust image and video dehazing with visual artifact suppression via gradient residual minimization,'' in Proc. Eur. Conf. Comput. Vis., 2016, pp. 576–591.

[28] Q. Zhu, J. Mai, and L. Shao, ''A fast single image haze removal algorithm singcolorattenuationprior,''IEEETrans.ImageProcess.,vol.24,no.11, pp. 3522–3533, Nov. 2015.

[29] H. Yang, C. H. Yang and Y. J. Tsai, "Y-Net: Multi-Scale Feature Aggregation Network With Wavelet Structure Similarity Loss Function For Single Image Dehazing", 2020 IEEE International Conference on Acoustics Speech and Signal Processing (ICASSP), pp. 2628-2632, 2020.

[30] Y. Fan, Y. Cao and X. Yang, "Unmanned Aerial Vehicle Remote Sensing Image Dehazing via Global Parameters", 2020 IEEE International Conference on Artificial Intelligence and Information Systems (ICAIIS), pp. 349-354, 2020.

[31] R. W. Liu, J. Nie, S. Garg, Z. Xiong, Y. Zhang and M. S. Hossain, "Data-driven trajectory quality improvement for promoting intelligent vessel traffic services in 6G-enabled maritime IoT systems", IEEE Internet Things J., vol. 8, no. 7, pp. 5374-5385, 2021.

[32] R. W. Liu, W. Yuan, X. Chen and Y. Lu, "An enhanced CNN-enabled learning method for promoting ship detection in maritime surveillance system", Ocean Engineering, vol. 235, no. 109435, 2021.

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